# Part I Introductory Topics For Everyone

## 1) Introduction and Motivation

### 1) Online Controlled Experiments Terminology.

#Overview

"online Controlled Experiments Terminology" introduces standard vocabulary used throughout the book and in the practice of A/B testing (online experiments). Clear understanding of these terms is crucial for designing, analyzing and interpreting experiments.

☑ Core terminology (Explained Simply with Analogies + Math + Code).

1. Experiment (A/B Tests)

Definitions: A scientific tests where user are randomly split into two or more groups.

- control Group (A): Gets the original experience.
- Treatment Groups (B): Gets the modified/new experience

Analogy: like testing two types of fertilizers on similar plots of lands. Group A gets old fertilizers; group B gets new fertilizers. We compare plant growth

Math Concept: Random assignment ensures independent samples so we can validly compare means/metrics.

2. Treatment Effect

Definition: The difference in outcomes between control and treatment. This is the effect caused by the change.

Formula: Treatment Effect = $\bar{x}_{treatment} - \bar{x}_{control}$.

3. Randomization

Definition: Randomly assigning users to control or treatment groups to eliminate bias.

Analogy: Tossing a fair coin to assign users — no favoritism or skewed distribution.

Math Concept: Random assignment → balanced groups
↳ valid inference via CLT

4. Unit of Randomization / Analysis

Definition: The entity over which randomization is done and analysis is measured.

Common Units:
- user (most common)
- session (can bias metrics)
- Cookie / device (less stable).

Interlinked Concept: Bias and confounding if wrong unit is used.

Example: If a user clears cookies and is treated again, it breaks independence assumption.

5. Metric

Definition: A quantifiable measure used to compare performance between control and treatment.

Types:

- Invariant Metric: should not change (e.g. user count)
- Evaluation Metric: Measure success (e.g clicks, revenue)

Analogy: In a school exam experiment, invariant metric = number of students, evaluation metric = average score.

6) Null Hypothesis & Statistical Significance

Null Hypothesis ($H_0$): No difference between control and treatment.

Statistical Significance: How confident we are that the observed effect isn't due to chance.

Math Formula:

$$Z = \frac{\bar{X}_T - \bar{X}_C}{\sqrt{\frac{\sigma_T^2}{n_T} + \frac{\sigma_C^2}{n_C}}}$$

7) Power and Sample Size.

- Power: Probability that your experiment will detect a true effect.
- Same Size: Need to ensure experiment has sufficient power.

8. Lift

Definition: Percent increase (or decrease) in metric compared to control.

Formula: $\text{Lift} = \frac{\bar{X}_T - \bar{X}_c}{\bar{X}_c} \times 100\%$

9. Variance & Standard Error.

Variance: Spread of the data

Standard Error: Variability of the mean across samples.

Used for: Confidence intervals and hypothesis testing.

10. Interlocked Topics.

These terms are tied tightly with:

- Why Experiment? - Because of causality vs correlation.

10. P-value Misterpretation

Clarification: A low p-value doesn't prove the treatment is good - just that it is unlikely to observe the result by chance under the null.

DOMS | Page No.
Date / /

# ~~Chapter~~ (2) Why Experiment? Correlation, Causality and Trustworthiness.

→ This chapter explain why controlled experiments (like A/B tests) are essential in online environment to establish causal relationships - not just correlations. It addresses,

↳ Why observational data can be misleading

↳ Why randomization is the foundation for trustworthy experimentation

## Correlation ≠ Causation

→ Correlation: A statistical relationship between two variables.

↳ Example: Users who search more tend to buy more.

→ But: Maybe engaged user both search and buy more, rather than 'searching causing buying'

→ Observation alone can't prove causality because of confounding variables (like user engagement, seasonality, or geography)

# Controlled Experiments Enable Causal Inference

⇒ In a A/B test, you randomly assign users to
- Group A (control)
- Group B (treatment)

→ If the groups are large and randomly split, all other factors (confounders) are balanced on average.

→ Any statistically significant difference in outcome can be attributed to the treatment.

This is causal inference.

Analogy: Medicine Trial (Refined)

Goal: Determine if a new headache medicine really works.

Setup:
- Group A (control): Gets a sugar pill (placebo) ← no active ingredients.
- Group B (treatment): Gets the new headache medicine

☑ Key Rule: People are randomly assigned to ensure that both groups are similar in every way -- age, stress levels, hydration etc.

Observations

-> After 1 week, Group B reported fewer headaches than Group A.

:-> Since both groups were otherwise the same, we can confidently say:

▶ The medicine caused the improvement.

## How This Mirrors Online A/B Testing

| Medical Trial | A/B Test |
|---|---|
| Group A: Placebo | Group A: sees old website |
| Group B: New medicine | Group B: sees new feature |
| Random patient assignment | Random user assignment |
| Outcome: Headache reduction | Outcomes: Clicks, sign-ups, sales increase |
| Result: Drug caused the change | Result: Feature caused the improvement. |

Why This works:

-> Because of randomization, we've isolated the only difference between groups -- treatment itself.

→ so, any statistically significant change is causally linked to that treatment.

Core Principle
Controlled experiment are like clean lab test — they filter out noise, leaving you with clear cause-effect answers.

## Observational Studies Have Hidden Bias

Observational studies analyze data from real world behaviours without random assignment. While sometimes useful, they can mislead you about causality.

### What Can Go wrong?

1. Selection Bias
→ People exposed to the new feature may already differ from those who weren't.
→ Example: Power users get early access making the feature look more effective

2. Self-selection Bias
→ users may choose to use a feature, skewing the results.

→ Example: A "Try Before" button — only curious + tech-savvy users opt in, inflating success metrics.

3. Survivorship Bias.

→ You're only analyzing users who stuck around, ignoring those who churned.

→ Example: Measuring satisfaction only among long-term customers excludes unhappy one who left.

4. Seasonality or Time-Based Effects.

→ If one group experiences the features during a holiday sale or a bug, it confounds the result.

Why it's Problem

Even with advanced techniques like propensity score matching (try to match users on age, region, device, type etc). you're only adjusted adjusting for observed variables.

You can't control for unknown or unmeasured factors — like mood, intent, or competitor activity — which might be driving the

DOMS | Page No.
Date / /

## Quite Worthy Insights

> "No matters how sophisticated your regression is, it cannot match the power of a randomized experiment."
> — Kohavi, Tang, Xu.

**Bottom line:**
Observational studies can suggest a relationship, but only randomized controlled experiment can establish causality.

## Controlled Experiments Vs. Observational Study

### Controlled Experiment

→ You randomly assign half of users to see the discount (Group B), and the other half not see it (Group A).
→ Since users were assigned randomly, both groups are similar on average - same kinds of users, same days, same behaviour mix.
→ So, if Group B buys more, you can confidently say, "The discount caused the increase in purchase!"

DOMS | Page No.
Date / /

## Observational Study (No Random Assignment)

→ You compare users who chose to click the discount banner with those who didn't.
→ But users who click discount are already different - maybe they're more price-sensitive or frequent shopper.
→ Even if Group B buys more, you can't be sure why! ⇒ was it the discount or just who saw the banner?

| Features | OS | CE |
|---|---|---|
| How users are assigned | By behaviour or circumstance (not controlled) | Randomly, by system |
| Bias risk | High: different users may react differently | Low: group are balanced on average |
| Can prove causation? | No - can only suggest connection | Yes - can isolate the effect of a change |
| Confounding factors | Present (e.g. time, location, user type) | Minimized through randomisation. |
| Example | "User who clicked discount bought more" | "Users who saw discount (randomly) bought more". |

| Conclusion reliability | Weak - many possible hidden explanation | Strong - only one major difference (the change) |
|---|---|---|

→ Controlled experiments are like clinical trials for your website. They give you trustworthy causal results.

→ Observation studies are useful for generating ideas, but can't trust them for decision making.

Randomization = Trust.

What is Randomization?

→ Randomization means assigning users randomly into different groups (control vs. treatment), without regard to their:

- Age
- Behavior
- Location
- Device
- Browsing history.

DOMS Page No.
Date / /

## Why Randomization Build Trust?

Because the groups are randomly created:

→ Any confounding factors (like time of day, user demographic, user intent) are balanced on average.

→ So, any difference in outcomes is very likely due to the treatment itself (the change you're testing)

## Example: Website Button Color Test

- You test red vs. blue buttons for signups
- Randomly assign 50% of users to red, 50% of to blue.
- After the experiment:
  - Red group has a 7% signup rate
  - Blue group has a 5% signup rate.

⇒ Since users were randomly assigned, you can trust that the red button caused the increase -- not that "more motivated user saw red".

Analogy: A/B Test

New feature vs. old version

Randomly assigned users.

Outcomes: Did users convert, click, stay longer?

→ Above case, random assignment ensures that the only meaningful difference between groups is what they received.

No Randomization ? No Trust

Randomized Controlled Trials (RCTs) are considered the gold standard in:

without randomization

-> You don't know if users who saw the new button were more likely to convert anyway.

-> You can't separate what caused the outcome from who the user were

DOMS | Page No.
Date / /

# Why Randomisation = Gold standard?

Randomised Controlled Trials (RCTs) are considered the gold standard in:

- Medicine → Testing new drugs
- Public policy → Evaluating social programs
- Tech → Online A/B Testing

They are trusted because they:

→ Eliminate selection bias (on average)
→ Enable clear causal inference
→ Can be replicated and scaled.

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## ② Why Experiments? Correlations, Causality, and Trustworthiness &

# Goal of This Section.

To answer one big question:

| "Why do we need to run online experiment (like A/B tests) instead of just looking at data?"

### 1. Correlation: When Two Things Happen Together

# Definition

| Correlation means two thing change together - but we don't know why.

# Example

Let's say you see this in your app data:

- Users who spend more time on the site also buy more.

This is a correlation - two things go up together

But you don't know if:

- A: Time causes purchases
- B: Purchases cause time
- C: A third reason (like promotions) causes both.

### 2. Causation: When One Thing Causes Another.

# Definition

| causation means: "changing X directly change Y"

You need intervention - change something and measure the effect.

# Real-world Analogy

Imagine:

- You see that students who take online notes score higher.
- So you say: "Taking online notes causes better grades."

But what if:

- These students were already smarter?
- Or more motivated?

⇒ You don't trust that correlation. You must test it.

☑ Controlled experiment

- Randomly assign students to "take notes online" vs "papers only"
- Then compare their scores.

Now, you know whether the method causes better results.

3. Why Experiment?

| Because randomized experiment are the only trustworthy way to measure causality

☑ What Experiments Do:

1. Randomly assign users to control and treatment groups.
2. Apply your change (like a new button) only to treatment
3. Measure results.

If the treatment group performs better, you can say:
⇒ "This change caused the improvement".

# Trustworthiness.

- No bias: Random assignment makes both group equal on average.
- Fairness: You isolate only the change, not other effects.
- Reliable decisions: You can act based on real evidence.

## (3) Necessary Ingredients for Running Useful Controlled Experiments

The authors stress that just running an experiment isn't enough — to get useful, trustworthy results you need certain core ingredients.

1. Clear Hypothesis
2. Randomization.
3. Sufficient sample size (Statistical Power)
4. Reliable Instrumentation (Accurate Data Collection).
5. Relevant Metrics.
6. Test Duration (Run the test long enough to cover variations (day of week, seasonality, user habits).
7. Analysis plan

Decide beforehand how you'll analyze results, including:

- Hypothesis test type
- Significance threshold
- Segments to analyze.

Why?
Avoids cherry-picking results after the fact.

## 4. Tenets

The authors introduce a guiding principle ("Tenets") that every online experiments should follow. These are like rules of the road for A/B testing – they keep answers trustworthy, reproducible and actionable.

1. Trust the data - But Ouery Tools Properly Run Experiments
2. Speed Matters - Run Fast - Learn Fast.
3. Decentralize - let Teams Test Independently.
4. Evaluate long-term Impact, Not just short-term Goal.
5. Don't Overreact to Noisy Data.
6) Make Decisions Based on Pre-Defined Metrics.
7. Beware of Launching without Testing.
8. Understand Practical vs Statistical Significance.
9. Don't Stop Tests Early Just Because They look Good.

## 5. Improvement over Time

(1) Small changes. Big Gain Over Time

(2) Compound Effect of Iteration.

If each experiment increases a metric by even 1%, repeated over multiple cycles, the cumulative effect is substantial.

(3) Speed and Feedback loop.

- Insight: Rapid experimentation accelerates Learning. Shorter cycle times lead to more tests – and more opportunities for improvement.

- connections:
- ~~Insight~~:
- Tactics: "speed Matters" encourages just this iterative flow.
- Necessary: Requires good instrumentation in products, metrics, and automation to support rapid.

(4) Avoiding Over-Reliance on Big Bets (Changes).

(5) Examples of Interesting Online Controlled Experiments is

(1) What The's unit Covers

→ strategy refers to the overarching goal and direction- what you ultimately want to achieve.

→ Tactics are the specific actions or experiments you perform to move toward that strategic goal.

(2) Strategy vs. Tactics: Definitions & Analogy.

- Strategy
  - Big-picture aims (e.g. "Increase coverage order value by 10% this quarter")
  - Determines which metrics matter most.
- Tactics
  - Concrete experiments (e.g. "Test a free-shipping banner at checkout")
  - Measured via A/B tests to check their contribution toward strategic goals.

# Real-world analogy

Strategy is the destination ("Climb Mount Everest")
: tactics are the steps (pick route, pack gear, train).
Experiments help test each tactic's effectiveness before committing.

# Math Concept: Strategy focuses on OEC, Tactics Deliver lift.

In the book, authors recommend defining an Overall Evaluation Criterion (OEC) - a single metric tied to strategy - and the running multiple experiments (tactics) that together optimize it.

If each tactic yields a small attributable lift $\Delta_i$, over n tactics

$$\text{Total lift} \approx \sum_{i=1}^{n} \Delta_i$$

Experiment quantifies each $\Delta_i$, so strategy is evidence-driven.

(7) Additional Reading 24 ✓.